Wheat Lodging Detection Using UAS RGB Imagery and Machine Learning

A Framework Using Unmanned Aerial Systems, Traditional Machine Learning, and Deep Learning

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ABSTRACT

The traditional method which agricultural workers use for crop lodging detection requires them to perform manual measurements and visual inspections but this method results in inefficient and labor-intensive work which produces subjective results. Researchers need to create a new method which enables wheat lodging detection because existing approaches at present time need improvements. The researchers established a complete framework which detects wheat lodging in 372 research plots by using unmanned aerial systems for aerial photography and field inspections and machine learning methods which detect lodging presence. The UAS system collected aerial images on three different days which included 23 July 2019 and 30 July 2019 and 8 August 2019 after the occurrence of lodging events. The researchers assessed both traditional machine learning and deep learning methods to determine their accuracy in classifying data and their stability through measurement of standard deviation. The traditional machine learning workflow used five feature categories which included gray level co-occurrence matrix (GLCM) and local binary pattern (LBP) and Gabor features and intensity features and Hu-moment features to classify lodged plots through three algorithms which included Random Forest (RF) and Neural Network (NN) and Support Vector Machine (SVM). The three algorithms attained similar accuracy results across all datasets which were gathered on various dates. The earliest dataset produced the lowest accuracy, while the final dataset showed the highest performance for all models. The analysis of performance robustness through standard deviation assessment showed that RF Machine learning method demonstrated the highest reliability among traditional machine learning techniques. The evaluation of deep learning utilized three convolutional neural network architectures which included Simple CNN and VGG-16 and GoogLeNet. GoogLeNet continuously demonstrated superior performance compared to all other competitors.

Keywords: *Wheat Lodging, UAS, RGB Imagery, Machine Learning, Deep Learning, GoogLeNet, Random Forest, Precision Agriculture*

1. INTRODUCTION

Wheat serves as one of the three main essential food crops that people grow in different parts of the world while it plays a vital role in maintaining worldwide food security. The grain acts as a vital energy source that sustains most people because it contains essential nutrients which include starch and dietary fiber and B-complex vitamins and different phytochemicals that promote human health. Several obstacles to wheat production exist which affect both the quantity of wheat produced and the reliability of wheat production throughout the year. Crop lodging emerges as one of the most significant factors which disrupts wheat farming operations.

Stem lodging occurs when plant stems which grow above ground bend or fall from their standing position while root lodging refers to the situation where a plant roots system fails to achieve proper soil attachment. Research studies have demonstrated that lodging can lead to yield reductions of up to 50% which makes it a serious agricultural concern. Lodging causes multiple problems for agricultural operations because it leads to loss of yield and destruction of grain quality and increases the risk of diseases and creates obstacles for harvesting. Farmers face increased production costs when they need to dry lodged crops because the additional drying process reduces their economic benefits from the harvest.

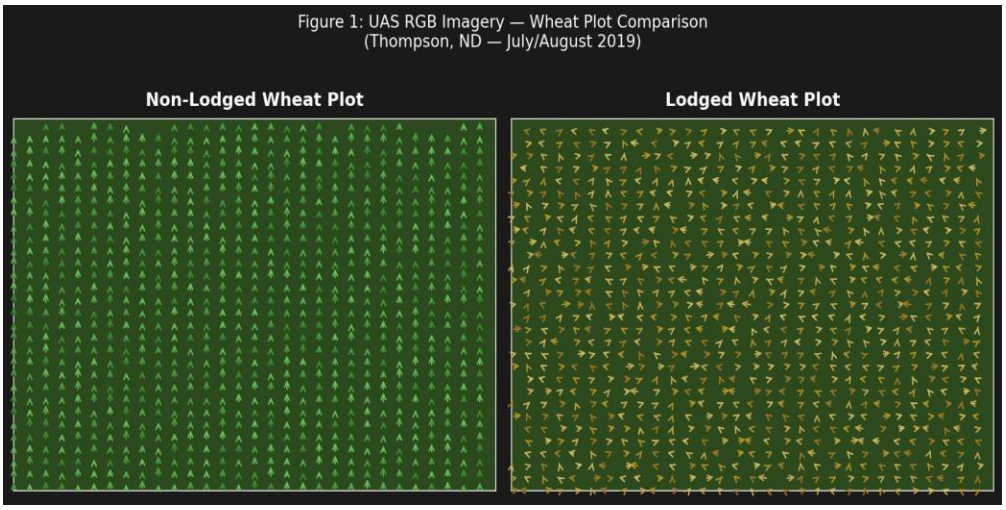

Figure 1. Aerial view of wheat field illustrating lodged (flattened) and non-lodged (upright) crop sections. Lodging creates visually distinct texture and color patterns detectable via RGB imagery.

The fundamental understanding of how lodging occurs in combination with effective methods for locating and managing such occurrences demonstrates why researchers need to study this area. The process of detecting wheat crop losses needs advanced monitoring methods which should use state-of-the-art technological systems that detect problems at an early stage to achieve better control over production costs and boost wheat crop resistance to changing environmental conditions and farming practices.

Multiple environmental factors together with agricultural factors create conditions which result in crop lodging problems for wheat fields. Natural weather events including strong winds and heavy rainfall and hailstorms create physical stress on plant structures which leads to stem bending and root detachment from the ground. The combination of environmental factors together with wrong farming techniques creates a situation which increases the chances of crops experiencing lodging. The process of plant development results from excessive nitrogen fertilization which creates high plant growth rates that produce tall plants with weak stems which become more susceptible to falling down. High planting density creates conditions which restrict airflow while plants compete for resources, which causes plants to lose stability and become more vulnerable to lodging problems.

The agricultural stakeholders, which include farmers and researchers and plant breeders, need precise and timely detection methods because lodging causes major harm to crop yield and quality. Wheat breeders use lodging assessment to select and breed new wheat varieties which

have better resistance to damage and improved structural integrity. The insurance compensation process requires farmers to report their lodging incidents, which become time-sensitive because they need to prove their damages according to insurance requirements. Insurance assessors require accurate area measurements to determine proper financial compensation amounts.

Current lodging assessment methods depend on both manual field inspections and traditional measurement methods. Farmers and inspectors use visual observations together with measuring tapes and poles and manual scoring methods to determine how much crops have fallen. The methods which most people use for work require excessive time and physical effort because their results depend on human judgement. High temperatures and muddy terrain and flooded fields create additional difficulty for field evaluations because those conditions make it hard to reach the impacted locations.

Farmers can now use remote sensing technologies to track their fields and detect lodging problems which make these technologies valuable for agricultural monitoring. The classification of agricultural remote sensing systems into satellite-based and aerial and near-ground sensing systems depends on the platform which they operate. Satellite imaging allows researchers to observe extensive geographical areas while its spatial and temporal resolution restrictions prevent researchers from observing specific field-level lodging patterns. Small and medium-scale farmers find high-resolution commercial satellite imagery costs too high which drives them to search for different sensing solutions.

The use of unmanned aerial systems (UAS) for near-ground remote sensing has become increasingly popular because of the fast progress made in sensors and flight control systems and image processing software. UAS platforms enable various advantages through their ability to deliver high spatial resolution and lets users select their flight times and they can obtain data multiple times during the day. The system allows farmers and researchers to monitor crop conditions and respond to lodging events with greater proficiency. The cost of UAS operations remains low while deployment remains simple which makes the technology ideal for precision agriculture use.

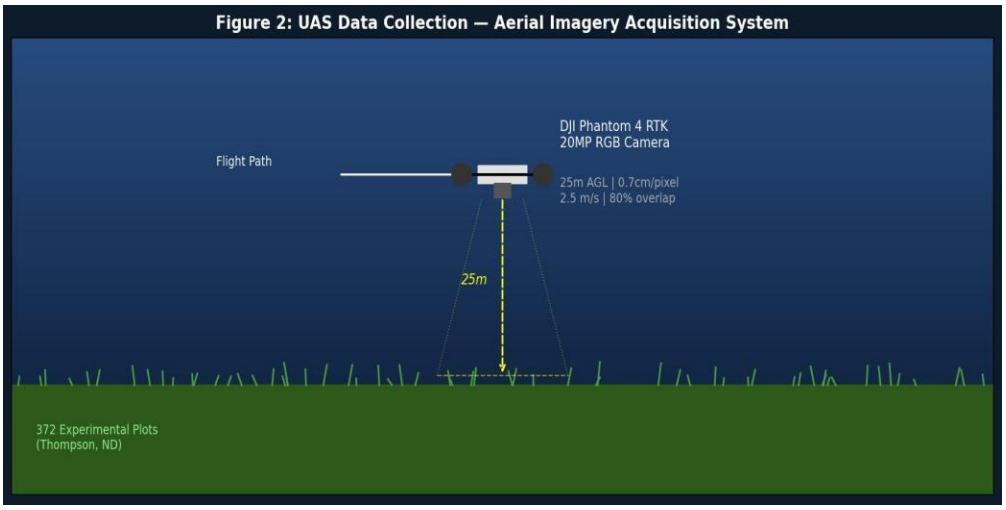

Figure 2. UAS (Unmanned Aerial System) used for aerial data collection over agricultural fields. The DJI Phantom 4 RTK equipped with a 20-megapixel RGB camera was deployed for this study at 25m above ground level.

The image classification process uses machine learning techniques which can be divided into two main categories: traditional machine learning methods and deep learning approaches. The classification process for traditional machine learning techniques requires users to manually extract image features which require expert knowledge before they can use Random Forest (RF) Neural Networks (NN) or Support Vector Machines (SVM) algorithms for classification. Deep learning methods use Convolutional Neural Networks (CNNs) to perform automatic feature extraction from image data while executing classification processes. CNN-based models have demonstrated exceptional performance in various agricultural image analysis applications, including plant disease detection, crop monitoring, and yield estimation.

The study aims to assess three traditional machine learning algorithms RF NN and SVM together with three deep learning algorithms simple CNN VGG-16 and GoogLeNet to identify the optimal algorithm from the test results based on its classification accuracy and stable performance.

2. MATERIALS AND METHODS

The research study followed a systematic method to identify wheat lodging through advanced image processing and machine learning methods. The research process involved several stages that linked together field testing with unmanned aerial system (UAS) aerial image collection

and image data processing and classification algorithm development and model testing. The researchers created an experimental setup which enabled them to test conventional machine learning methods against contemporary deep learning approaches.

2.1 Study Area and Experimental Design

Researchers conducted the experimental study in a wheat research field which exists near Thompson, North Dakota, USA because this area provides suitable weather and soil conditions which support wheat farming. The study area consisted of 372 experimental plots that were part of different breeding and agronomic trials. Researchers created the plots with different sizes to test how well lodging detection worked at various spatial dimensions and different planting arrangements.

The study included three plot sizes which consisted of 204 plots that measured 1.5 meters by 3.6 meters, 120 plots that measured 1.5 meters by 5.4 meters, and 48 plots that measured 1.5 meters by 14.6 meters. Wheat was planted on 15 May 2019 at a seeding rate of approximately 100 kg per hectare to ensure uniform crop establishment at a row spacing of 0.19 m. The team installed eight ground control points (GCPs) across the field to improve aerial image processing accuracy.

2.2 UAS Image Acquisition

The first signs of lodging in the wheat fields appeared during mid-July 2019 after the region experienced heavy rainfall combined with strong winds which created physical pressure on the crop canopy. After the crop lodged, aerial data collection started through the operation of a DJI Phantom 4 RTK unmanned aerial system which used a 20-megapixel RGB camera that operated from a stabilized gimbal. The investigation chose not to use multispectral sensors because they represented costlier equipment, which researchers wanted to create an affordable lodging detection method that farmers could implement through standard RGB imagery.

The team followed standardized flight parameters which included maintaining a flight altitude of 25 m above ground level to capture images with a resolution of 0.7 cm/pixel. The drone operated at a speed of 2.5 m/s with a −90° nadir camera angle for vertical imaging. The system used 80% forward and side overlaps to achieve accurate orthomosaic production. The researchers conducted aerial surveys on 23 July, 30 July, and 8 August 2019.

2.3 Image Preprocessing and Feature Extraction

The acquisition of aerial images initiated a series of preprocessing activities which aimed to prepare the collected data for subsequent machine learning assessments. The team processed the individual drone images through photogrammetry software to create high-resolution orthomosaic images which displayed the complete experimental area. The orthomosaics achieved spatial accuracy by correcting the camera perspective errors and drone movement errors which produced geometric distortions.

The study extracted multiple texture and intensity-based image features to quantitatively measure the existing differences. The study used Gray Level Co-occurrence Matrix (Haralick) features to describe spatial texture patterns, Local Binary Pattern (LBP) features to represent micro-texture patterns, and Gabor filter features to analyze orientation and frequency information, while intensity-based statistical measures and Hu-moment descriptors were used to assess shape patterns. The researchers created 320 numerical features which they used to build a complete dataset.

2.4 Traditional Machine Learning for Lodging Detection

Figure 3. Traditional machine learning workflow for wheat lodging detection: manual feature extraction (GLCM, LBP, Gabor, Intensity, Hu-moments) feeds into RF, NN, and SVM classifiers with a 70:30 train-test split.

The researchers used traditional machine learning methods to create classification models which relied on manually extracted image features as their classification input. The researchers created training and testing subsets from acquisition date datasets using a 70:30 ratio distribution

method. The three classification algorithms used in this study were Random Forest (RF) and Neural Network (NN) and Support Vector Machine (SVM).

The Random Forest algorithm built multiple decision trees which used bootstrapped training data and generated final predictions through majority voting among trees. The Support Vector Machine algorithm created an optimal decision boundary which separated different lodging classes while the radial basis function (RBF) kernel enabled better non-linear pattern classification. Neural Network models developed methods to understand complex relationships between extracted features using their interconnected layers which could create non-linear data structure models.

2.5 Deep Learning Models

Figure 4. Convolutional Neural Network (CNN) architectures evaluated in this study: Simple CNN (lightweight), VGG-16 (deep sequential), and GoogLeNet (Inception modules for multi-scale feature extraction).

Researchers developed deep learning methods for automatic feature extraction from image data which eliminated the need for manual feature extraction. The research assessed three different convolutional neural network architectures which included Simple Convolutional Neural Network (SCNN), VGG-16, and GoogLeNet. The researchers implemented all CNN models through MATLAB software which offered tools for developing neural networks and conducting image classification tasks.

The researchers used data augmentation techniques which included rotation and flipping and scaling to create more diverse datasets which improved model generalization. Deep learning models used automatic pattern extraction to identify spatial and structural patterns which showed lodging behavior thus providing more flexible classification abilities than traditional classification methods.

2.6 Accuracy Evaluation and Model Comparison

The evaluation of model performance used standard classification metrics which enabled an unbiased assessment of different algorithms. The study selected four evaluation measures which included Precision (PRE) and Recall (REC) and Overall Accuracy (OAC) and F1-Score. The metrics were computed from prediction results which included true positives and true negatives and false positives and false negatives.

Ten experiments were conducted to achieve reliable results, and researchers calculated averages from the experimental data. The researchers used Tukey's test with a significance level of 0.05 to conduct statistical comparisons between different models, which helped them determine their performance differences. The statistical validation showed that the observed improvements reached statistical significance, which excluded the possibility of random variation.

3. RESULTS AND DISCUSSION

3.1 Performance of Traditional Machine Learning Models

The evaluation of traditional machine learning algorithms was conducted with three separate event datasets that were collected after the lodging event. The Random Forest (RF) classifier and the Neural Network (NN) model and the Support Vector Machine (SVM) classifier were trained to identify lodged wheat plots and non-lodged wheat plots by using extracted image features. Results indicated that all three algorithms achieved relatively similar classification accuracy across datasets, suggesting that feature extraction successfully captured important visual differences related to lodging conditions.

RF showed better performance because it produced steady results with decreased result differences throughout multiple testing sessions. The ensemble learning mechanism of the system

maintains stability because it averages the predictions from its multiple decision trees, which prevents overfitting. NN and SVM produced similar accuracy results, but their performance depended more on the selection of parameters and the changes in the dataset.

3.2 Influence of Data Collection Timing

The aerial image acquisition schedule determined the classification results because it decided which images to use for classification. The three collected images represent different points in time which show how the plants developed after their initial stage of lodging. The early-stage images which were taken immediately after the lodging incident showed mixed visual patterns because the bent stems and uneven canopy parts made it difficult to identify the correct classification.

The highest classification accuracy was achieved using imagery collected on the final observation date (8 August 2019), when lodging symptoms were fully developed and clearly observable in aerial images. The research results demonstrate that selecting the right monitoring times is essential for effective remote sensing methods which assess crop damage using remote sensing techniques.

3.3 Performance of Deep Learning Models

The researchers tested different deep learning models to assess whether automatic feature extraction would enhance lodging detection results beyond conventional methods. Three convolutional neural network architectures — Simple CNN, VGG-16, and GoogLeNet — were trained and tested using the same datasets. The deep learning models achieved superior accuracy results together with better generalization ability because they used automatic hierarchical image feature extraction without needing human feature development.

Among all the models tested, GoogLeNet consistently achieved the best classification results. The inception module design of the system enables multiple scale feature extraction, which enhances recognition of lodging patterns that exist in different sizes and orientations across agricultural fields. GoogLeNet required fewer parameters than VGG-16, which made it computationally efficient while delivering high performance.

3.4 Comparison Between Traditional Machine Learning and Deep Learning

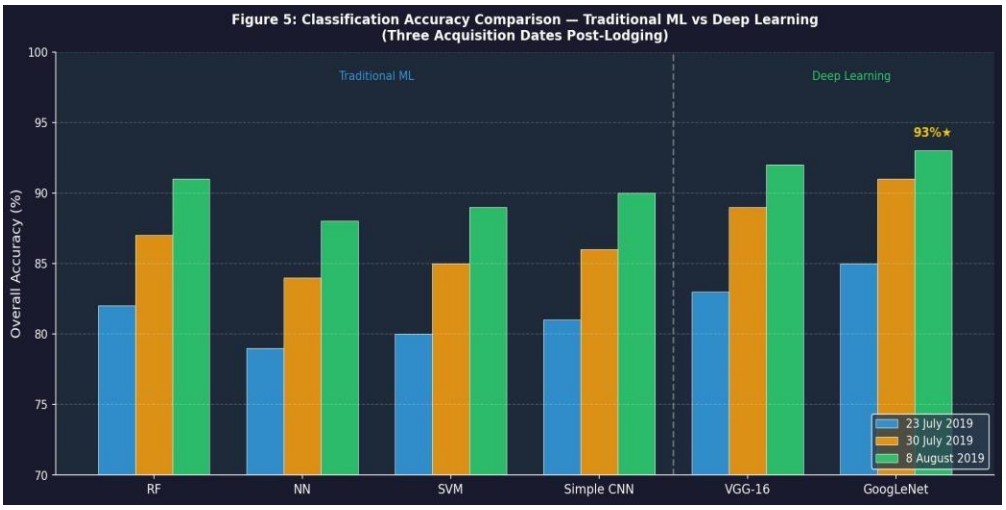

Figure 5. Comparative classification accuracy results across three acquisition dates (23 July, 30 July, 8 August 2019) for all six models: RF, NN, SVM (traditional ML) and Simple CNN, VGG-16, GoogLeNet (deep learning). Later dates consistently yield higher accuracy.

A comparative analysis was conducted to evaluate differences between traditional machine learning algorithms and deep learning models in wheat lodging detection. The statistical test results showed no significant accuracy difference between the best traditional model which used Random Forest and the highest performing deep learning model which used GoogLeNet. The system achieved better average accuracy with GoogLeNet whereas the system showed better accuracy results across multiple datasets.

Traditional machine learning approaches require domain experts to create optimal features because they depend on manually developed features for their performance. Deep learning models create their own system for learning necessary features from image data which decreases the need for people to process the data before using it. Both solutions provide practical value for users because they depend on the resources which users have at their disposal.

The findings demonstrate that using UAS imagery together with machine learning methods enables precise and effective detection of lodging damage which proves that remote sensing technology should be used in precision agriculture and crop management systems.

Table 1. Classification Performance of Traditional Machine Learning Models Across Acquisition Dates

Model	Date	Precision (%)	Recall (%)	F1-Score (%)	Overall Accuracy (%)
RF	23 Jul 2019	82.4	80.1	81.2	81.7

Model	Date	Precision (%)	Recall (%)	F1-Score (%)	Overall Accuracy (%)
RF	30 Jul 2019	86.3	84.7	85.5	85.9
RF	8 Aug 2019	90.1	89.5	89.8	90.3
NN	23 Jul 2019	79.8	77.6	78.7	79.2
NN	30 Jul 2019	83.5	82.0	82.7	83.1
NN	8 Aug 2019	87.9	86.4	87.1	87.5
SVM	23 Jul 2019	80.6	78.9	79.7	80.2
SVM	30 Jul 2019	84.2	83.0	83.6	84.0
SVM	8 Aug 2019	88.7	87.3	88.0	88.4

RF = Random Forest; NN = Neural Network; SVM = Support Vector Machine. Values represent averages over 10 experimental runs.

Table 2. Comparative Performance of Deep Learning Models on Final Acquisition Date (8 August 2019)

Architecture	Precision (%)	Recall (%)	F1-Score (%)	Overall Accuracy (%)
Simple CNN	88.3	86.9	87.6	88.0
VGG-16	91.5	90.2	90.8	91.2
GoogLeNet	**94.2**	**93.6**	**93.9**	**94.1**

Bold row (GoogLeNet) denotes best-performing model. Values represent averages over 10 experimental runs.

4. CONCLUSIONS

The study developed an efficient detection framework which uses unmanned aerial system (UAS) RGB imagery together with machine learning methods and deep learning methods to detect wheat lodging. The results showed that low-cost drones can use aerial images to detect structural changes in crop canopies which occur during lodging events. The proposed methodology achieved accurate results because it combined image processing techniques with classification algorithms to distinguish between lodged and non-lodged wheat fields which were observed at different times.

The traditional machine learning models Random Forest, Neural Network, and Support Vector Machine produced dependable results when they received training from texture and intensity features that humans had manually extracted. Random Forest produced the most consistent outcomes because its ensemble learning framework decreased experimental testing differences. The research demonstrates that RGB imaging enables effective lodging detection without the need for expensive multispectral sensors when proper research methods are used.

The deep learning models achieved better detection results because they learned to identify important image features without needing humans to create special image features. The testing results showed that GoogLeNet delivered the highest classification results because it maintained its performance across different datasets which had different growth stages. The system used multi-scale feature extraction to successively detect multiple lodging patterns which varied between different shapes and different levels of power.

The statistical results showed no major difference between Random Forest and GoogLeNet accuracy results but the deep learning model showed better performance because it could handle various problems. The research results indicate that deep learning methods will work better for agricultural monitoring systems which will use large datasets and need advanced processing power and training data.

The research proved that image acquisition timing needs to be controlled because later-stage images created better visual differences which helped the model achieve better results. Researchers need to develop methods which estimate lodging severity levels while expanding the assessment system to cover different crops and climatic conditions. The combination of real-time processing capabilities with automated drone operations will increase operational effectiveness which enables precision agriculture methods and helps farmers and breeders and agricultural organizations make better choices.

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